



— URBAN ANALYTICS · ACTIVE SCHOOL TRAVEL

Can children **walk to school?** Singapore's school-travel friendliness

A full replication of [Land 2024, 13\(8\):1319](#) (Lanzhou, 151 primary schools) for **160 Singapore primary schools** — scoring the quality of the 1 km walking catchment around each school from space syntax, pedestrian safety, greenery and footpath provision, then asking **what drives it** with a Geographic Detector.

160

PRIMARY SCHOOLS

1 km

NETWORK CATCHMENT

52

HIGH FRIENDLINESS

53

LOW FRIENDLINESS

235k

SPACE-SYNTAX NODES

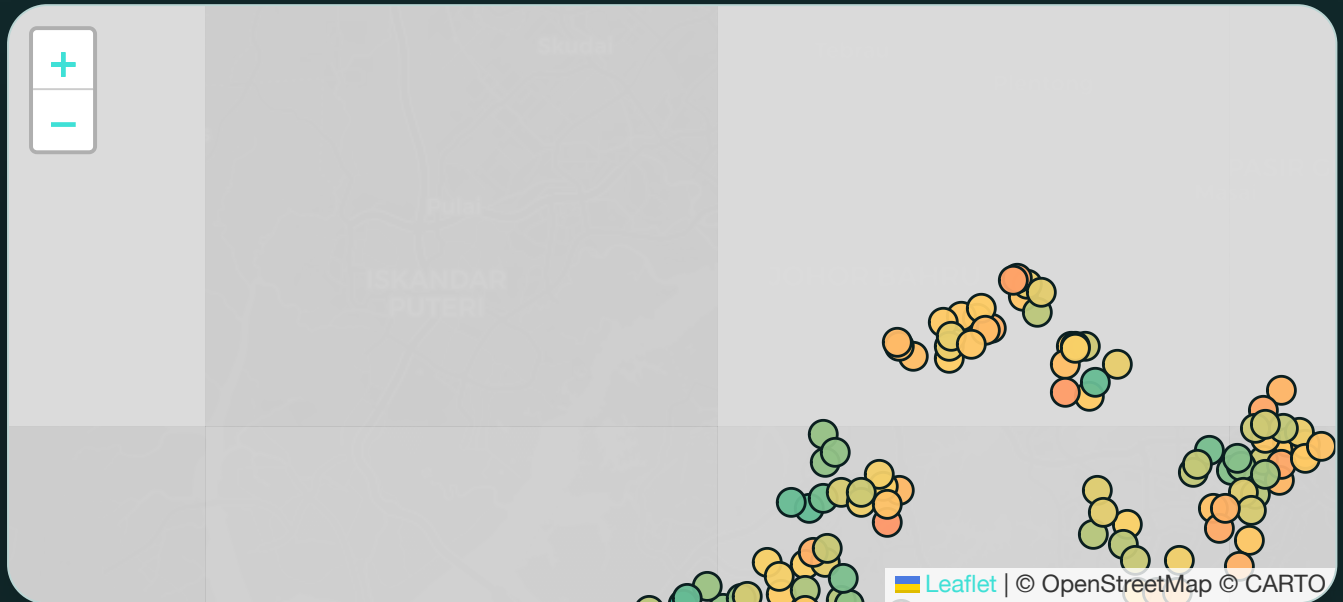
REPLICATING

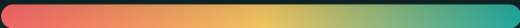
“Evaluating the Quality of Children’s Active School Travel Spaces and the Mechanisms of School District Friendliness Impact Based on Multi-Source Big Data”

Lu, C., Yu, C. & Liu, X. (2024). Land 13(8), 1319 · MDPI · doi:10.3390/land13081319 · 151 primary schools, Lanzhou, China · [MDPI](#)

The map

Each school at its location, coloured by the friendliness of its walking catchment.
Click a dot for the breakdown.



Less friendly  More friendly

Catchment = 1 km network distance · MOE home-school priority band

Key terms, in plain language

This study chains six methods: **schools** → **walking network** → **space syntax** → **catchment** → **entropy index** → **Geographic Detector**. Here is what each one means, the data behind it, and why it matters for planning.

01 **Schools** *where every journey ends*

The government primary schools, each pinned to its exact location. Everything in the study is measured in the area around these points — they are the anchor of the analysis.

IN THIS STUDY 160 of Singapore's ~179 MOE primary schools (a few dropped as data duplicates); spread West 43 · North-East 42 · Central 28 · North 28 · East 27.

FOR PLANNERS MOE's Primary-1 registration gives priority to homes within 1 km (then 1–2 km), so the area right around each school is where most of its pupils actually live — and could walk.



02 **Walking network** *the web of footpaths*

A digital map of every footway, sidewalk, park path and pedestrian crossing, stitched into one connected graph the computer can “walk” along. Built from OpenStreetMap.

IN THIS STUDY 170,121 nodes (junctions) and 463,880 edges (path segments) covering all of Singapore.

FOR PLANNERS This is the real surface children travel on. Distances are measured **along paths**, not straight lines — so a canal or expressway blocking a route is correctly counted as a detour, the way a child experiences it.



03

Space syntax *how readable the streets are*

An established urban-morphology science that scores a street layout on two things — **Integration** (how easy a place is to reach from everywhere else) and **Choice** (how likely a street is to lie on a natural through-route). Simple, well-integrated networks are easy to navigate; tangled ones are not.

IN THIS STUDY Computed with the cityseer engine on a 235,600-node “dual” model of the network, at 800 m and 1600 m walking radii.

FOR PLANNERS A legible network is one a 7-year-old can follow without getting lost, and where people and traffic move predictably. It is a structural wayfinding-and-safety measure, independent of how nice any single street looks.

**04**

Catchment *the active school travel space*

The bundle of streets a child can reach within **1 km of walking along the network** from the school gate — not a 1 km circle on a map, but the real reachable area. This is the “Active School Travel Space” the study evaluates.

IN THIS STUDY 160 catchments; on average ~31 km of walkable streets over ~0.9 km² each, from compact (3.8 km of streets) to sprawling (66 km).

FOR PLANNERS It is drawn to match the MOE 1 km priority band exactly, so it is the precise area where an upgrade would change the most pupils' walk to school.



05

Entropy index *one fair score from six measures*

The six raw measures (Integration, Choice, crossings, signals, greenery, footpath density) are fused into a single **0–100 score** by the **entropy weight method** — a statistical rule that gives more weight to indicators that genuinely vary and carry information, and less to flat ones. No official chooses the weights, so there is no agency bias.

IN THIS STUDY

Data-derived weights: greenery 0.26 · choice 0.19 · signals 0.19 · integration 0.16 · crossings 0.12 · footpaths 0.08. Schools then split into Low / Medium / High thirds.

FOR PLANNERS

An auditable, defensible score whose weights come from the data, not opinion — important for any public-facing prioritisation that has to withstand scrutiny.



06

Geographic Detector *what explains the map*

A spatial-statistics test that asks “how much of the friendliness pattern does each background factor explain?”, producing a power score **q** from 0 to 1. Its interaction mode checks whether two factors together explain more than either alone.

IN THIS STUDY

School centrality $q=0.22$ (strongest) · population density 0.11 · district size 0.08 · transport convenience 0.06 — all positive. Every factor pair is “nonlinear-enhancing” (density × centrality reaches $q=0.40$).

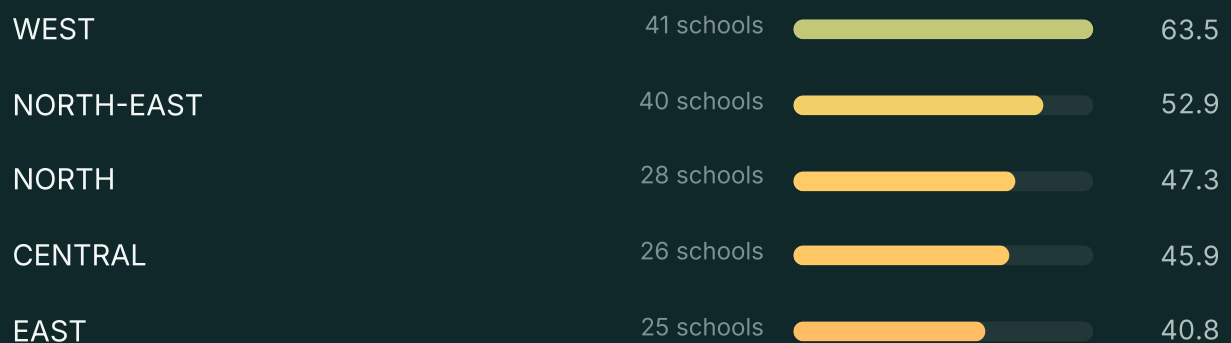
FOR PLANNERS

It reveals the **levers**: friendliness is not random — it tracks how centrally a school sits in the street network, amplified by density. That points to where structural change, not cosmetics, would help most.

A core-periphery pattern

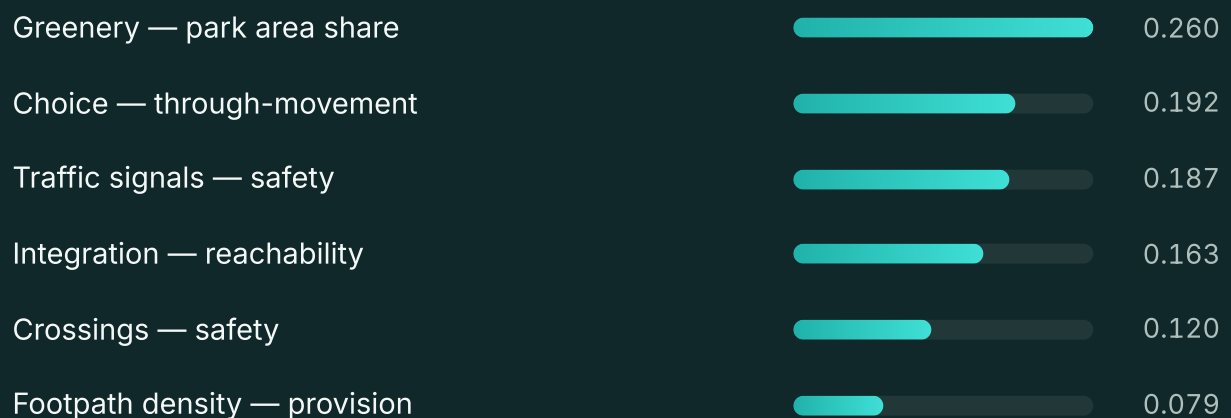
Friendliness rises toward the **newer planned towns** of the west and north-east and falls in the **dense central core** — the inverse of 15-minute amenity access. It reflects the walking **environment** (greenery, legible street networks, controlled crossings), the same core-periphery signature the Lanzhou study reported.

Mean friendliness by region



What makes a street friendly?

Objective entropy weights across all 160 catchments.



Most & least friendly catchments

Top 10

	SCHOOL	REGION	SCORE
1	Jurong West Pri Sch	WEST	100
2	Blangah Rise Pri Sch	CENTRAL	99
3	Angsana Pri Sch	EAST	99
4	West Grove Pri Sch	WEST	95
5	Frontier Pri Sch	WEST	91
6	Chua Chu Kang Pri Sch	WEST	85
7	Concord Pri Sch	WEST	84
8	Northland Pri Sch	NORTH	84
9	Westwood Pri Sch	WEST	84
10	South View Pri Sch	WEST	81

Bottom 10

	SCHOOL	REGION	SCORE
1	Raffles Girls' Pri Sch	CENTRAL	0
2	St Anthony's Canossian Pri Sch	EAST	6
3	East Spring Pri Sch	EAST	8
4	Bukit Timah Pri Sch	CENTRAL	13
5	Park View Pri Sch	EAST	16
6	Opera Estate Pri Sch	EAST	19
7	New Town Pri Sch	CENTRAL	20
8	Nanyang Pri Sch	CENTRAL	21
9	Bukit Panjang Pri Sch	WEST	22
10	Henry Park Pri Sch	CENTRAL	22

Reading the scores — what friendliness means here

The score is a **0–100 relative ranking** of how supportive each school's 1 km walking catchment is for a child travelling on foot — **100 = friendliest catchment in Singapore, 0 = least**. It rates the walking **environment**, not how close amenities are, and is comparative across the 160 schools rather than an absolute pass/fail.

WHAT IT CAPTURES

The child's walk

A high score means short, legible routes (few confusing turns), dense controlled crossings, leafy shaded streets and continuous footpaths — the conditions under which a parent lets a child walk or cycle to school unaccompanied.

WHY IT MATTERS IN SG

Policy-relevant by design

Singapore's MOE gives **admission priority within 1–2 km** of a school, so a large share of pupils live in genuine walking range. The score shows where that walking environment actually delivers — directly informing active-travel, road-safety and the “car-lite” agenda.

CONTEXT

Tropical & safety lens

In Singapore's climate, **greenery** (the heaviest-weighted indicator, 0.26) is shade and heat relief, not just amenity; **crossings & signals** are the front line of child road safety; **space-syntax legibility** means routes a young child can navigate. That is why these dominate the index.

HOW TO USE IT

Targeting, not judgement

A **Low** score is not a “bad school” — it flags a catchment where greening, crossing upgrades or footpath continuity would most improve children's walk. Bottom-decile catchments are the priority list for intervention.

Why the pattern inverts 15-minute access. Friendly catchments cluster in *newer towns* (Jurong West, Punggol) built with wide green park-connector footpaths and regular signalised crossings; the *dense central core* (Bukit Timah, Newton) scores lower because mature, fine-grained street networks have more turns, less green corridor and busier roads — even though shops sit closer. The score is measuring a different thing than proximity: the *quality of the journey*, which is what determines whether a child walks.

What drives friendliness — the mechanisms

Geographic Detector q-statistic: the share of friendliness variance explained by each contextual driver (held strictly **out** of the index to avoid circularity). All four push friendliness up.

Single-factor power (q)

School centrality	(+)		0.222
Population density	(+)		0.109
School-district size	(+)		0.084
Transport convenience	(+)		0.060

School centrality is the strongest single driver — schools embedded in well-connected street networks have friendlier catchments.

Interaction detector

FACTOR PAIR	Q	MAX SINGLE	TYPE
Population density n School centrality	0.395	0.222	nonlinear-enhance
School-district size n School centrality	0.365	0.222	nonlinear-enhance
Transport convenience n School centrality	0.364	0.222	nonlinear-enhance
Population density n School-district size	0.254	0.109	nonlinear-enhance
Transport convenience n School-district size	0.202	0.084	nonlinear-enhance
Population density n Transport convenience	0.167	0.109	bi-enhance

Every pair is **nonlinear-enhancing**: factors explain far more together than alone — friendliness is a system effect, not a single lever.

Methodology

Faithful to the source paper's chain — **schools** → **walking network** → **space syntax** → **catchment** → **entropy index** → **Geographic Detector** — adapted to Singapore and computed end-to-end on Propheus infrastructure. The logic: define the real walkable area a child crosses to reach school, measure its **environmental quality** from objective open data, fuse those measures into one score with no subjective weighting, then statistically test which contextual factors **explain** the resulting pattern. Where Lanzhou used the Amap routing API and Baidu street view, Singapore substitutes the OpenStreetMap pedestrian graph and the Digital Atlas v4 layers; the street-view dimension is deferred to Phase 2.

Data sources

Primary schools (points)	Digital Atlas v4 places master — 160 schools after cleaning
Walking network	OpenStreetMap pedestrian graph — 170,121 nodes / 463,880 edges
Road crossings · traffic signals	OpenStreetMap — 40,551 crossings · 5,443 signals (safety)
Parks & green space	OpenStreetMap — 4,658 polygons (greenery)
Bus stops · MRT stations	OpenStreetMap — 5,938 bus · 324 rail (transport-convenience driver)
Population density	Digital Atlas v4 subzone population ÷ subzone area

STEP 1 · CATCHMENT

1 km network catchment

For each school, the streets reachable within **1 km network distance** (not Euclidean) on the OSM walk graph — the MOE home-school registration priority band. The corridor = reachable edges buffered 40 m.

STEP 2 • SPACE SYNTAX

Angular Integration & Choice

cityseer angular analysis on the **dual graph** (235,600 nodes). Harmonic closeness → **Integration** (reachability); betweenness → **Choice** (through-movement), at 800 m & 1600 m radii.

STEP 3 • SAFETY & GREENERY

Objective environment

Controlled crossings & traffic signals per km of route (safety); park/green area as a share of the catchment (greenery); walkable footpath density (provision). All clipped to the catchment corridor.

STEP 4 • INDEX

Entropy-weighted composite

Six indicators min-max normalised, then combined by the **entropy weight method** — weights derived from each indicator's information content, no subjective tuning — and rescaled 0–100.

Formulae

Entropy weight of indicator j over n catchments:

$$p_j = r_{ij} / \sum_i r_{ij} \quad \cdot \quad e_j = -(1/\ln n) \sum_i p_{ij} \ln p_{ij} \quad \cdot \quad w_j = (1-e_j) / \sum_j (1-e_j)$$

Friendliness, and the Geographic Detector power statistic q (h = driver strata, 5 quantiles):

$$\text{Friendliness} = \sum_j w_j r_{ij} \quad \cdot \quad q = 1 - (\sum_h N_h \sigma_h^2) / (N \sigma^2) \in [0,1]$$

Disjoint sets. Index indicators (route environment) and Geographic Detector drivers (population density, transport convenience, district size, school centrality) are kept strictly separate, so q is never circular — the central design rule of the analysis.

Scope — Phase 1. This measures network structure plus objective environment proxies. The source paper's street-view experiential layer (green-view index, sky/enclosure, sidewalk ratio from semantic segmentation of Google Street View / Mapillary) is Phase 2 and requires a GPU. School list = 160 matched & cleaned from the v4 places master (vs MOE's 179). Computed entirely on Propheus servers from Digital Atlas v4 data + OpenStreetMap.



Powered by **Propheus Digital Atlas SGP v4.8**

Active School Travel Space friendliness · Singapore · 160 primary schools · entropy-weighted index · Geographic Detector. Method after Land 2024, 13(8):1319. Data: Digital Atlas v4 (places, population, boundaries) + OpenStreetMap (walk network, crossings, signals, parks, transit).